

Computer Science Notes

A Beginner-Friendly Guide to the Core Topics

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1. What is Computer Science?

Computer Science (CS) is the study of computers and computational systems: how information is represented, stored, processed and communicated, and how problems can be solved using algorithms and programs. It combines mathematics, logic and engineering.

Main areas of Computer Science

- **Programming and software development** - writing instructions that computers can run.
- **Algorithms and data structures** - efficient ways to solve problems and organise data.
- **Computer systems** - hardware, operating systems and networks.
- **Data and databases** - storing, retrieving and analysing data.
- **Artificial Intelligence** - machines that learn from data and make decisions.

2. Hardware and Software

Hardware is the physical part of a computer that you can touch. Software is the set of programs and instructions that tell the hardware what to do.

Component	Function
CPU (Central Processing Unit)	The brain of the computer; executes instructions.
RAM (Memory)	Temporary, fast storage for data currently in use. Cleared when power is off.
Storage (HDD / SSD)	Permanent storage for files and programs.
Motherboard	Main circuit board connecting all components.
GPU	Handles graphics and parallel computation, widely used in deep learning.
Input / Output devices	Keyboard, mouse, scanner (input); monitor, printer (output).

Types of software

- **System software** - operating systems, drivers, utilities.
- **Application software** - word processors, browsers, games.
- **Programming software** - compilers, interpreters, editors and IDEs.

3. Number Systems

Computers work with binary (only 0 and 1) because electronic circuits have two states: ON and OFF. A single binary digit is a **bit**; 8 bits make a **byte**.

System	Base	Digits used	Example
Binary	2	0, 1	1010
Octal	8	0 - 7	12
Decimal	10	0 - 9	10
Hexadecimal	16	0 - 9, A - F	A

Converting binary to decimal

Multiply each bit by 2 raised to its position (starting from 0 on the right) and add the results. For example, $1010 = 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 = 8 + 0 + 2 + 0 = 10$.

Units of data

1 Byte = 8 bits; 1 KB = 1024 bytes; 1 MB = 1024 KB; 1 GB = 1024 MB; 1 TB = 1024 GB.

4. Algorithms and Programming

An **algorithm** is a finite, step-by-step set of instructions to solve a problem. A good algorithm is correct, clear, finite and efficient.

Basic programming concepts

- **Variables** - named locations that store data.
- **Data types** - integer, float, string, boolean.
- **Conditionals** - if / else decisions.
- **Loops** - repeat actions (for, while).
- **Functions** - reusable blocks of code.
- **Object-Oriented Programming** - organising code using classes and objects (encapsulation, inheritance, polymorphism, abstraction).

Time complexity (Big-O)

Notation	Name	Example
O(1)	Constant	Accessing an array element by index
O(log n)	Logarithmic	Binary search
O(n)	Linear	Looping through a list
O(n log n)	Linearithmic	Merge sort
O(n ²)	Quadratic	Bubble sort

5. Data Structures

A data structure is a way of organising data so it can be used efficiently.

Structure	Description	Typical use
Array	Fixed-size, indexed collection	Storing lists of values
Linked List	Nodes connected by pointers	Dynamic insert / delete
Stack (LIFO)	Last in, first out	Undo feature, function calls
Queue (FIFO)	First in, first out	Task scheduling, printing
Tree	Hierarchical nodes	File systems, search trees
Graph	Nodes and edges	Maps, social networks
Hash Table	Key-value pairs with fast lookup	Dictionaries, caches

Common algorithms

- **Searching:** linear search, binary search (needs sorted data).
- **Sorting:** bubble, selection, insertion, merge and quick sort.

6. Operating Systems

An operating system (OS) manages hardware and software resources and provides services to programs. Examples are Windows, Linux, macOS, Android and iOS.

Main functions of an OS

- **Process management** - creating, scheduling and ending processes.
- **Memory management** - allocating and freeing RAM.
- **File management** - organising files and directories.
- **Device management** - controlling hardware through drivers.
- **Security and user management** - permissions and access control.

A **process** is a program in execution; a **thread** is a lightweight unit of execution inside a process.

7. Databases

A database is an organised collection of data. A Database Management System (DBMS) lets users create, read, update and delete data (CRUD).

Type	Description	Examples
Relational (SQL)	Data stored in tables with rows and columns, linked by keys	MySQL, PostgreSQL, SQLite
NoSQL	Flexible structures such as documents or key-value pairs	MongoDB, Redis

Key concepts

- **Primary key** - uniquely identifies each record.
- **Foreign key** - links one table to another.
- **Normalization** - organising tables to reduce redundancy.
- **Basic SQL commands:** SELECT, INSERT, UPDATE, DELETE, CREATE TABLE.

8. Computer Networks and the Internet

A computer network connects two or more devices to share data and resources.

Network type	Coverage
PAN	Personal area, a few metres (Bluetooth)
LAN	Local area, such as a home or office
MAN	Metropolitan area, a city
WAN	Wide area, such as the Internet

Important terms

- **IP address** - unique address of a device on a network.
- **DNS** - converts domain names into IP addresses.
- **HTTP / HTTPS** - protocols for web communication (HTTPS is encrypted).
- **TCP/IP** - the core protocol suite of the Internet.
- **Router** - forwards data between networks; **Switch** - connects devices inside a LAN.

9. Artificial Intelligence and Machine Learning

Artificial Intelligence (AI) is the ability of machines to perform tasks that normally need human intelligence. **Machine Learning (ML)** is a subset of AI where models learn patterns from data instead of being explicitly programmed. **Deep Learning** uses multi-layer neural networks and is a subset of ML.

ML type	How it learns	Examples
Supervised	From labelled data	Spam detection, image classification, price prediction
Unsupervised	Finds patterns in unlabelled data	Customer clustering
Reinforcement	Learns by rewards and penalties	Game-playing agents, robotics

Typical ML workflow

- Collect and clean data
- Split into training and test sets
- Train a model
- Evaluate it (accuracy, precision, recall)
- Deploy it, for example through an API

10. Cyber Security

Cyber security is the practice of protecting systems, networks and data from attacks.

The CIA triad

- **Confidentiality** - only authorised users can access data.
- **Integrity** - data stays accurate and unaltered.
- **Availability** - systems and data are accessible when needed.

Common threats and protections

- **Threats:** malware, phishing, ransomware, denial-of-service attacks.
- **Protections:** strong and unique passwords, two-factor authentication, firewalls, antivirus software, regular updates, data encryption and backups.

11. Quick Revision Points

- Computer Science studies how to represent, process and communicate information using algorithms.
- Hardware is physical; software is the set of instructions.
- Computers use binary; 8 bits = 1 byte.
- An algorithm is a step-by-step solution; Big-O measures its efficiency.
- Stack = LIFO, Queue = FIFO.
- The OS manages processes, memory, files and devices.
- SQL databases use tables; NoSQL databases use flexible structures.
- DNS maps domain names to IP addresses; HTTPS is the secure version of HTTP.
- AI contains ML, which contains Deep Learning.
- The CIA triad: Confidentiality, Integrity, Availability.

Prepared as study notes. Good luck with your studies!